

The manuscript presents a clearly structured and potentially useful system-integration framework for object-conditioned haptic teleoperation. Coordinating slave-side impedance, master-side haptic settings, and gripper-execution parameters through an interpretable strategy layer is relevant. The asynchronous perception architecture and the five-mode comparison are also thoughtfully presented. However, the current experimental design and statistical analysis do not yet provide sufficiently robust evidence for the principal claims. In particular, the very small participant sample, participant-level pseudoreplication, absence of online perception logs, and lack of channel-level ablation experiments substantially limit interpretation of the reported improvement.

1. The Friedman and Wilcoxon analyses use 27 matched blocks as the repeated-measures units, although these blocks are nested within only three operators. Blocks generated by the same operator are not statistically independent. Similarly, the matched-block bootstrap resamples the 27 block differences without preserving participant-level clustering and is therefore likely to underestimate uncertainty. Stating that the results are “conditional on the observed sample” does not resolve this dependence. The authors should either:
 - recruit a substantially larger number of operators and analyse the results using an appropriate participant-level mixed-effects or hierarchical model; or
 - restrict the analysis to descriptive within-sample findings and remove inferential population-level language and p-values.

A participant-clustered bootstrap or permutation analysis could be considered, but three clusters are insufficient for reliable inference. This is the most important limitation of the study.

2. Each operator completed 45 formal trials, while the five mode orders were preassigned but not fully counterbalanced. Consequently, mode, repetition, object allocation, learning, and fatigue may be confounded. A mode performed later could benefit from task familiarity, whereas a mode performed earlier could be disadvantaged.

The authors should provide the full trial sequences, analyse completion time against trial order and repetition, and demonstrate that mode C was not systematically favoured by order. A fully counterbalanced or randomized crossover design with more operators would be preferable.

3. The controlled visual test used fixed viewpoint, background, lighting, and known object instances, obtaining 100% classification accuracy. However, event-level class labels, confidence values, trigger timestamps, and trigger-to-contact timestamps were not retained during the human experiment. Therefore, it cannot be verified that the intended strategy was correctly and consistently activated in the trials from which the performance conclusions were drawn.

Because visual triggering is a central component of the proposed framework, the authors should report, for every online trial:

- detected class and selected strategy;
- confidence and false-trigger rate;
- camera-to-detection and detection-to-parameter-update latency;
- parameter-update time relative to first contact;
- cases in which no update or a fallback update occurred.

Without these data, the experimental results cannot be confidently attributed to the proposed object-conditioned perception–configuration pipeline.

4. Mode C changes both the master-side haptic channel and the gripper-execution channel relative to mode E. The observed 1.79-s difference therefore cannot establish whether the improvement arose from haptic feedback, gripper speed/force, their interaction, or true cross-channel coordination.

At minimum, the experimental design should include:

- impedance plus adaptive haptics, with fixed gripper parameters;
- impedance plus adaptive gripper parameters, with fixed haptics;
- the complete three-channel configuration.

A factorial or component-wise ablation study is necessary to substantiate the claim that coordinated multi-channel configuration is responsible for the performance improvement.

5. The initialized vector is reported as

$$\Theta_0 = [150, 10, 1.0, 0.5, 0.3, 0.10, 20]^T$$

but this is not identical to the balanced strategy, whose haptic dead zone, gripper speed, and requested force are 0.4 N, 0.05 m/s, and 15 N, respectively. In fact, the initialized gripper parameters correspond more closely to the stability-priority settings.

Thus, mode E does not compare complete configuration with a neutral or balanced impedance-only configuration. It compares mode C with an impedance-adapted system retaining potentially mismatched haptic and gripper settings. This could inflate the apparent benefit of mode C. The baseline selection must be justified, and additional comparisons using genuinely neutral or object-matched baseline settings should be included.

6. The strategy settings were chosen through “engineering screening,” but essential details are missing:
 - the number and ranges of candidate parameter combinations;
 - the operators, objects, and trials used during screening;
 - whether screening data were independent of the formal experiment;
 - the quantitative acceptance and exclusion criteria;
 - whether experimenters knew the intended hypotheses;
 - sensitivity of the conclusions to nearby parameter values.

If the same objects or operators contributed to parameter tuning and evaluation, the reported results may reflect task-specific optimization. The authors should document the complete tuning protocol and provide a sensitivity or robustness analysis.

7. Objects are manually categorized as fragility-priority, balanced, or stability-priority, but their mechanical properties, friction, allowable deformation, grasp-force limits, and slip thresholds are not measured. The paper cup, for example, is described as highly deformation-sensitive but is assigned to the balanced rather than fragility-priority strategy.

The class-to-strategy mapping currently functions as a manually designed lookup table rather than a general object-conditioned method. The authors should provide objective assignment criteria based on measurable properties or interaction requirements. Ideally, representative compliance, friction, deformation, and grasp-stability measurements should be reported.

8. The requested gripper force F_g is not measured fingertip force, while the haptic input is derived from the robot’s internal external-force estimator. No direct contact-force, object-deformation, slip, realized gripper-force, or realized master-force measurements are presented. Failure labels were assigned by direct observation without video recording or blinded reassessment.

Consequently, the study demonstrates a difference in task completion time but does not objectively demonstrate gentler contact, safer handling, improved grip stability, or reduced deformation. These claims should either be substantially narrowed or supported through synchronized force sensing, gripper aperture measurements, deformation tracking, video-based slip detection, and independently rated outcomes.

9. The reported median supervisory-cycle duration is 5.07 ms, which is slightly longer than the nominal 5-ms period of a 200-Hz loop. This raises questions about missed deadlines and timing jitter. The authors should report the actual loop-frequency distribution, maximum and percentile execution times, deadline-miss rate, operating-system scheduling, queue age, and computational load.

Moreover, 200 Hz is relatively low for high-fidelity haptic rendering. The relationship between the application loop and the Omega.7 internal servo loop should be clarified. Formal stability or passivity analysis is particularly important because impedance parameters and force-feedback gains change during operation.

10. The manuscript states that the baseline force-feedback command was transmitted without additional application-layer saturation and that saturation status was not recorded. This is a safety and reproducibility concern. The authors should report:
 - the Omega.7 hardware and software force limits;
 - maximum commanded and realized forces;
 - whether device-level saturation occurred;
 - rate limits or filtering applied to force commands;
 - the stability implications of immediate haptic-parameter changes.

Smoothstep interpolation of impedance values alone does not demonstrate stability of the complete bilateral system. A formal stability/passivity argument or experimental stability assessment over the parameter range is needed.

11. A single detection with confidence ≥ 0.25 permanently locks the strategy for the trial. The threshold is low and its selection is not justified. The framework should be tested under misclassification, distractors, occlusion, multiple objects, unknown classes, changing illumination, and object-instance variation. Threshold sensitivity should be reported, and the one-shot rule should be compared with alternatives such as multi-frame consensus, temporal filtering, confidence hysteresis, or post-contact verification. The safety implications of assigning an unmapped object to the balanced strategy also require discussion.
12. The study uses six known object classes, one instance or a very limited set of instances per class, prescribed placement, fixed orientation, one grasp-and-transfer task, and a manually specified mapping. This supports a semi-structured class-to-parameter lookup system, but not yet a generally applicable approach to heterogeneous-object teleoperation.
The authors should clarify what is fundamentally novel relative to existing object-conditioned impedance selection and shared-control methods. The title, abstract, and conclusions should explicitly limit the findings to predefined objects, strategies, and operating conditions unless experiments with unseen instances, new objects, environmental variation, and additional tasks are added.
13. All participants were right-handed but were required to operate the Omega.7 with their left hand because of the device placement. This unusual constraint may increase workload and variability and could interact with mode-dependent haptic feedback. The rationale should be justified, and ideally the platform should be configured for participants' dominant hand.
The authors should also define the task initiation and terminal events precisely, quantify initial object-position variation, and report phase-specific times for approach, grasp, transfer, and release. Phase-level analysis would help explain where the reported 1.79-s improvement originated.
14. Completion time improved, but the trajectory-length confidence interval included zero, success differences were not statistically clear, and pause-count findings were descriptive. Therefore, broad statements about improved "performance" should be replaced by the narrower finding of reduced completion time within the observed sample.
Failed trials were retained in the timing analysis, and the successful-only analysis was post hoc. The authors should explain how failure affects the common terminal endpoint and consider a prespecified composite measure such as successful completion time, penalized time, or time-to-success.
15. The literature survey on haptic feedback is incomplete, missing advanced techniques – e.g., A transparent teleoperated robotic surgical system with predictive haptic feedback and force modelling, A force-feedback methodology for teleoperated suturing task in robotic-assisted minimally invasive surgery, Real-time haptic characterisation of Hunt-Crossley model based on radial basis function neural network for contact environment, An actuated force feedback-enabled laparoscopic instrument for robotic assisted surgery, A review of recent methods and applications of virtual fixtures in robot-assisted surgery, Master-slave robotic system for needle indentation and insertion.
16. Clarify the exact physical meaning and units of

$$D_{API} = 2\zeta\sqrt{K}.$$

Since K contains both translational and rotational stiffness terms and no Cartesian inertia matrix is included, the dimensional interpretation is unclear. The exact Franka API implementation should be documented.

17. The sign convention for the impedance equation should be stated explicitly by defining whether the pose error is desired minus actual or actual minus desired.
18. The aperture cue u_g depends on the Omega.7 master-gripper angle rather than slave-gripper aperture, grasp force, or object deformation. It therefore appears to be a self-generated master-side cue rather than feedback about the remote grasp. The terminology and rationale should be revised, and its perceptual benefit should be separately tested.
19. Changing K_f and d together changes both feedback gain and the effective external-force threshold d/K_f . Reporting this effective threshold would make the haptic settings easier to interpret.
20. Explain the implications of using a fixed gripper target width of 0 m and an outer tolerance of 0.080 m for objects with substantially different dimensions.

21. Report the complete Raw NASA-TLX dimensional scores rather than only the overall arithmetic mean. Individual dimensions may reveal whether time pressure, physical demand, or frustration explains the differences.
22. Provide confidence intervals or participant-level plots for all principal descriptive outcomes. Means and standard deviations calculated from four or five blocks per object should not be interpreted as object-level evidence.
23. The ethics statement should identify the institution that issued the exemption and provide the exemption reference number or written determination date.
24. “Data available on request” is insufficient for a study whose central conclusions depend on a small dataset. The anonymized trial-level data, trial order, analysis code, detector configuration, trained weights or training protocol, and parameter-setting files should be deposited in a public repository.
25. Several figures contain substantial visual information. Font sizes, annotation density, and readability should be checked at the journal’s final column width.